Analysis Report

AutoPrep

December 18, 2024

Contents

1 Dataset Overview

Number of samples	1309
Number of features	13
Numeric features	6
Categorical features	7

1.1 Target Distribution

Category	Value
0	809
1	500

Table 1: Target Distribution

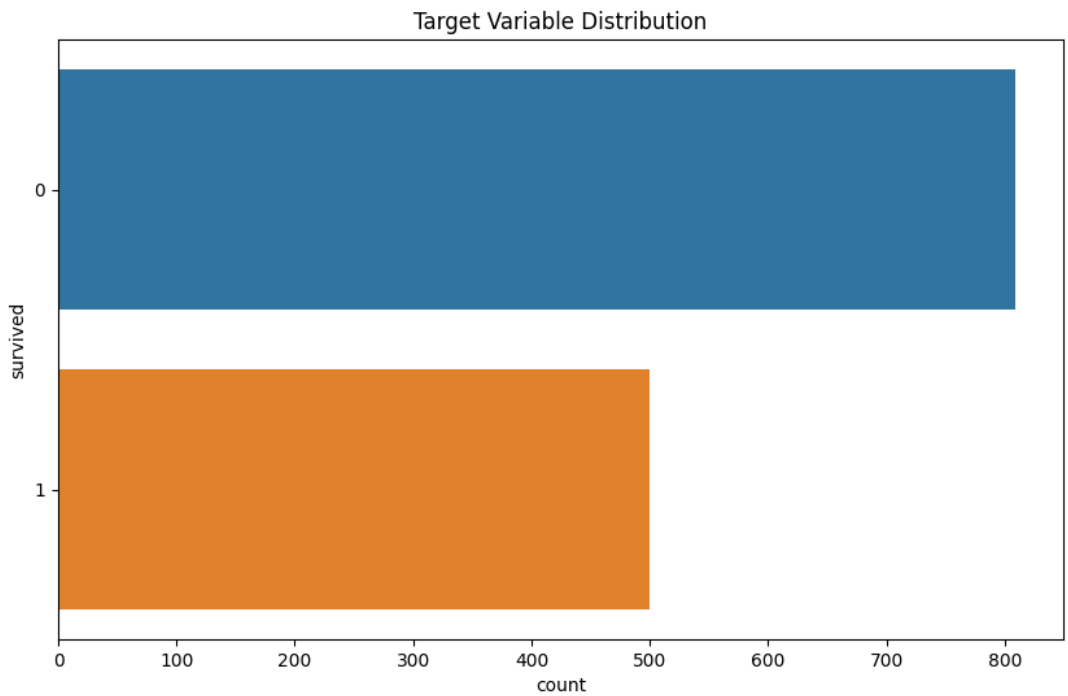

Figure 1: Target Variable Distribution

1.2 Missing Values

Category	Value
pclass	0
survived	0
name	0
sex	0
age	263
sibsp	0
parch	0
ticket	0
fare	1
cabin	1014
embarked	2
boat	823
body	1188
home___dest	564

Table 2: Missing Values

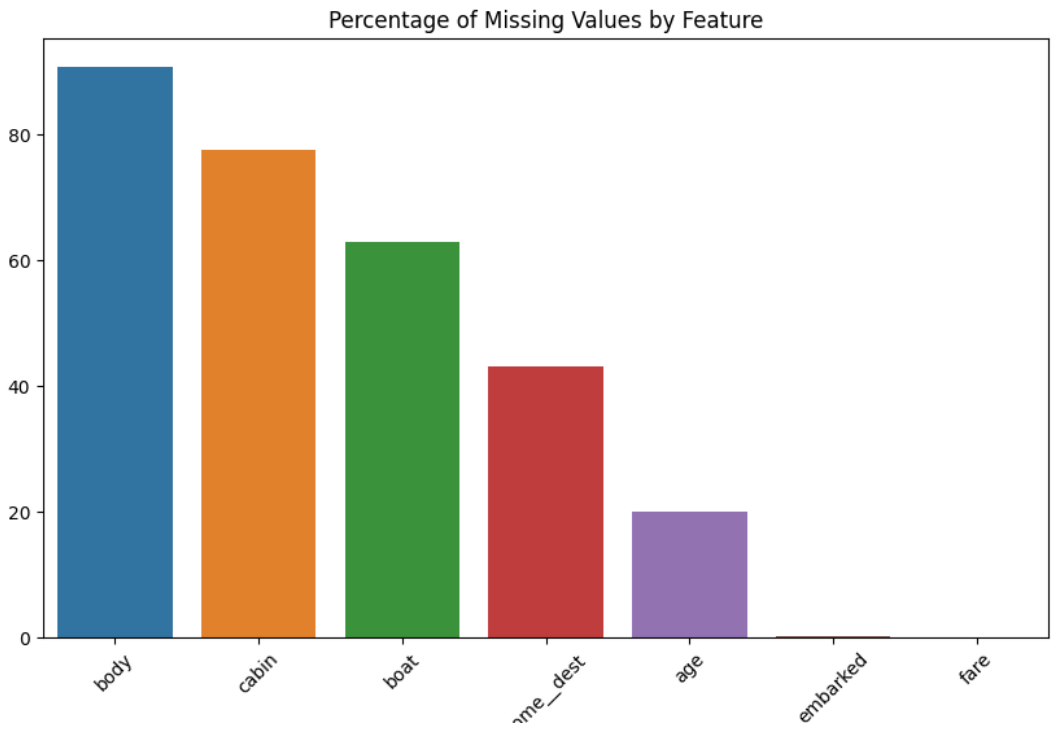

Figure 2: Missing Values Analysis

2 Exploratory Data Analysis

Visual analysis of the dataset features.

2.1 Numeric Features

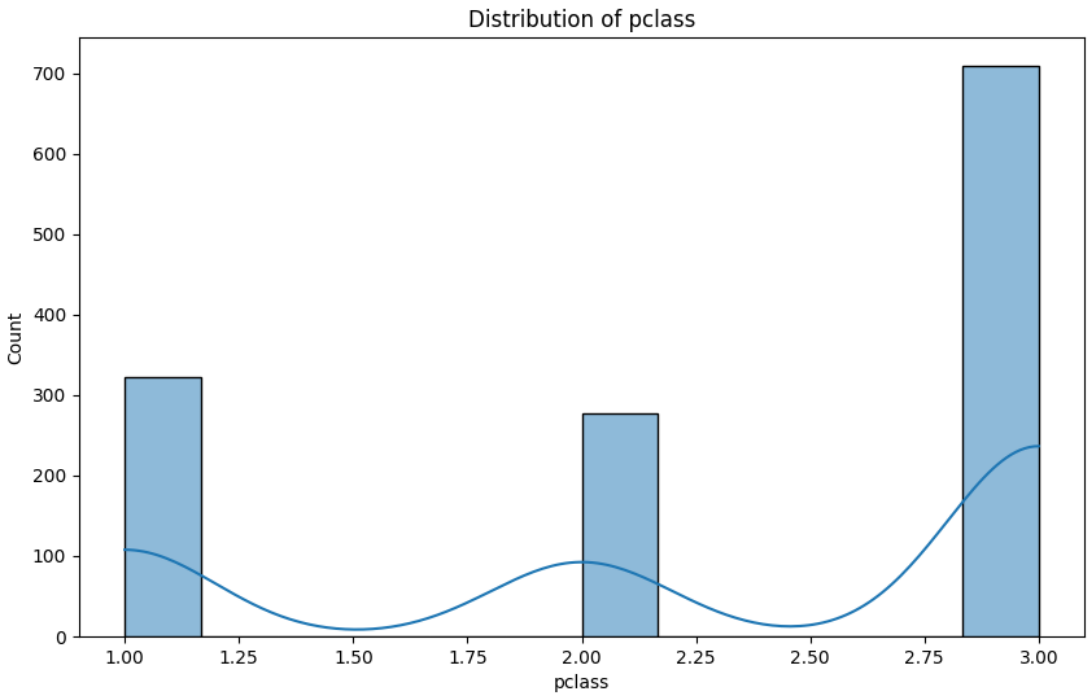

Figure 3: Distribution of pclass

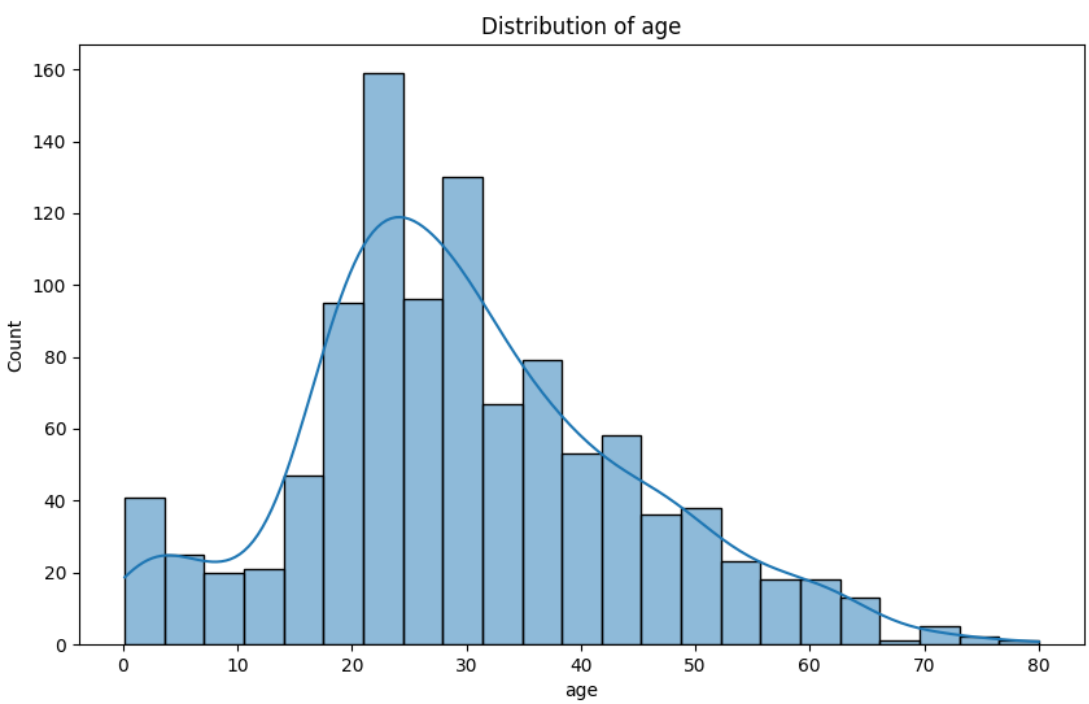

Figure 4: Distribution of age

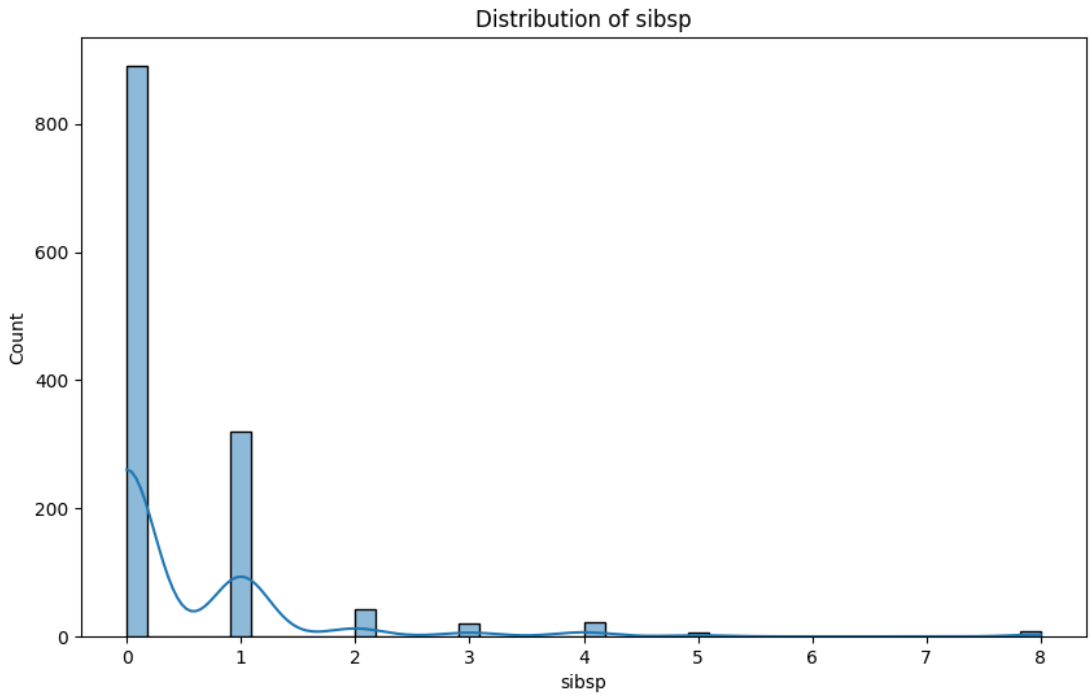

Figure 5: Distribution of sibsp

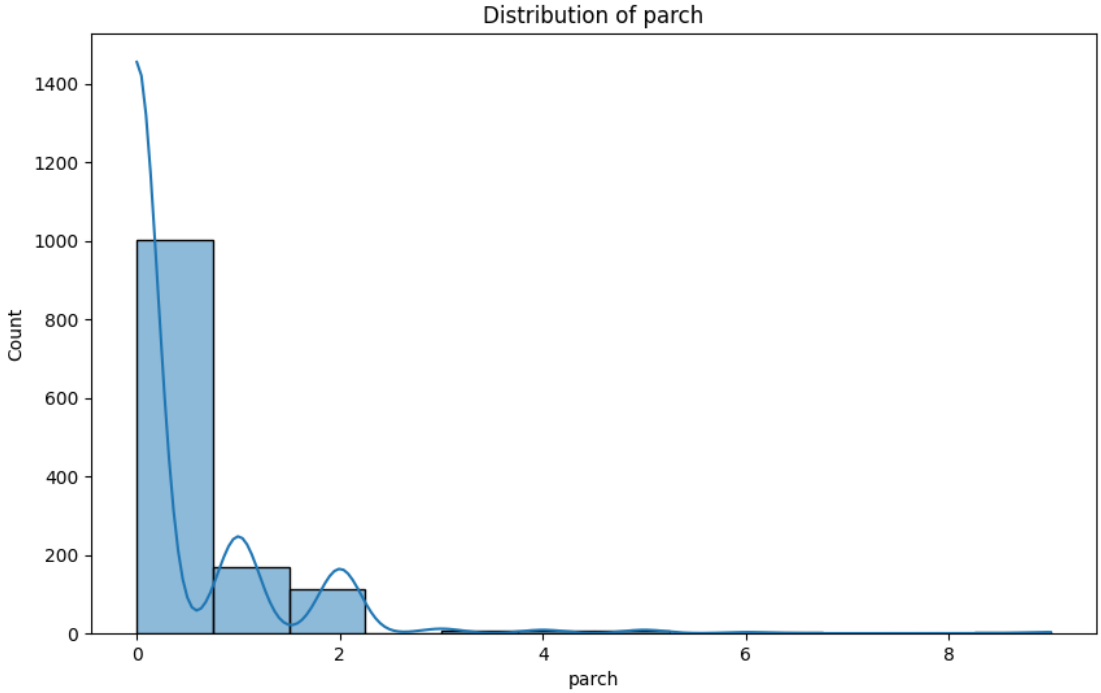

Figure 6: Distribution of parch

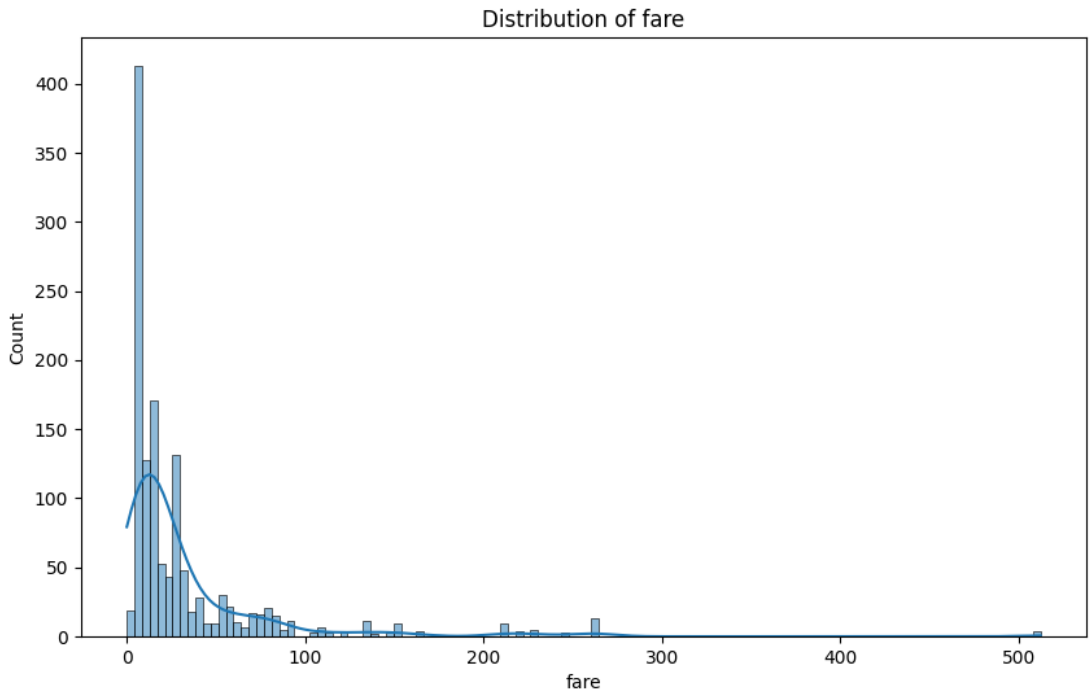

Figure 7: Distribution of fare

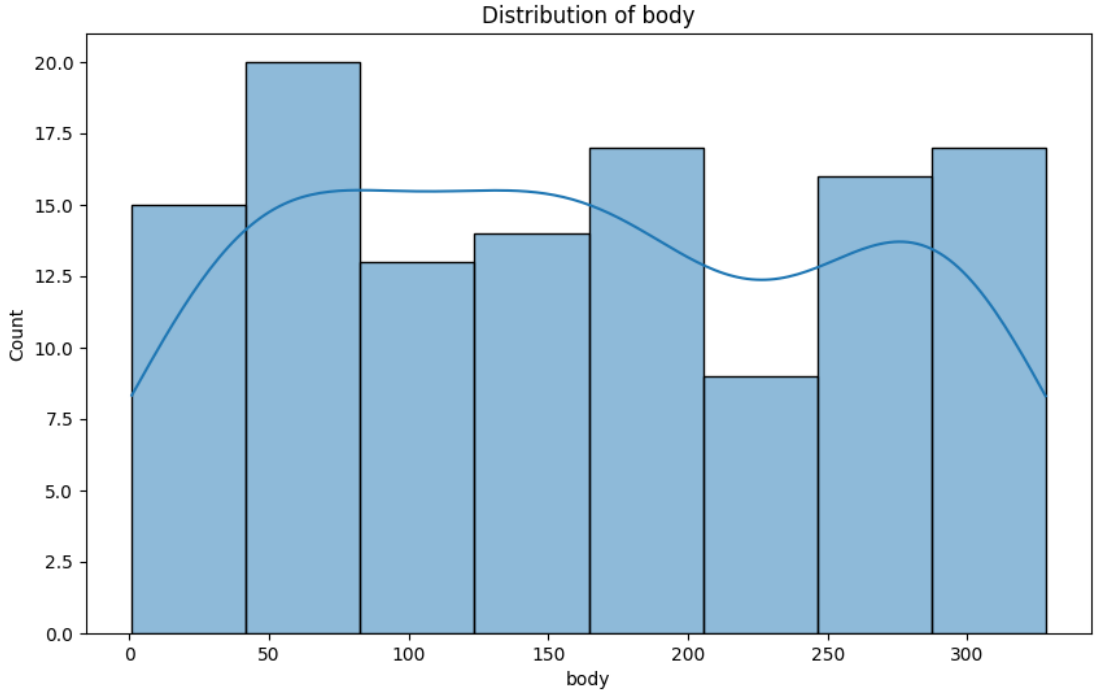

Figure 8: Distribution of body

2.2 Categorical Features

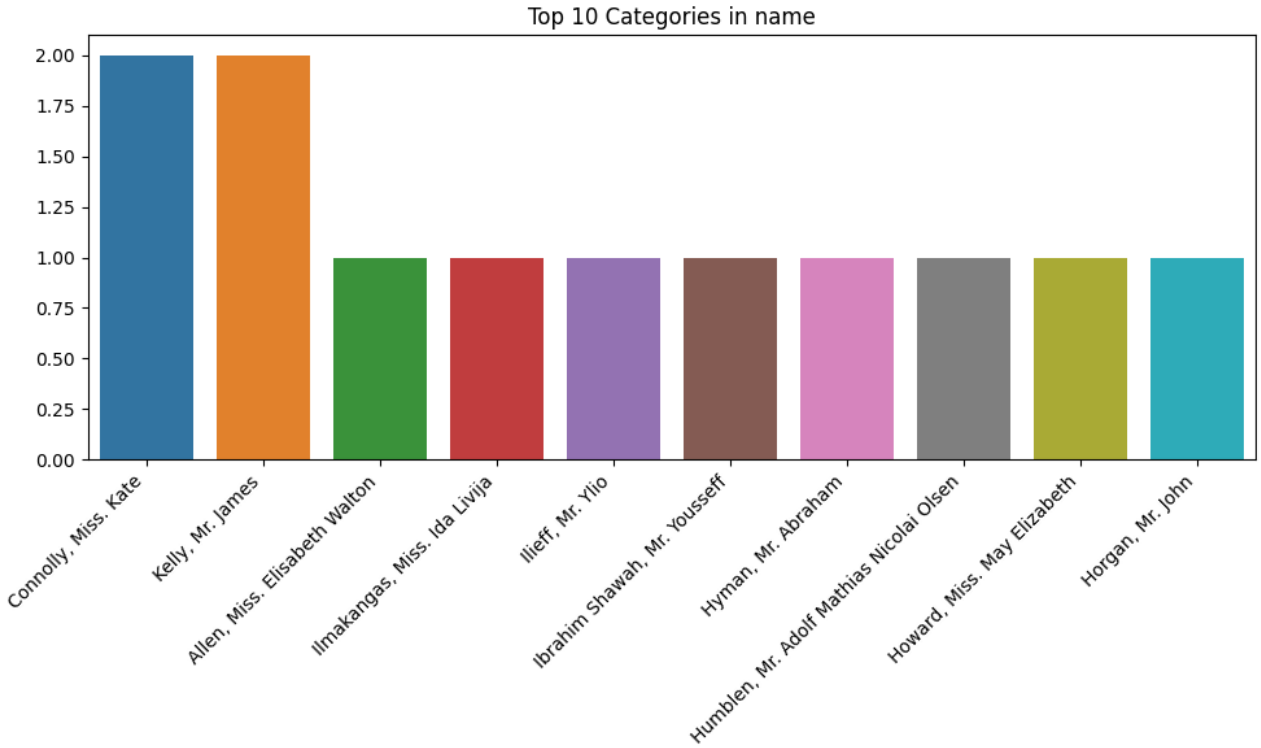

Figure 9: Distribution of name

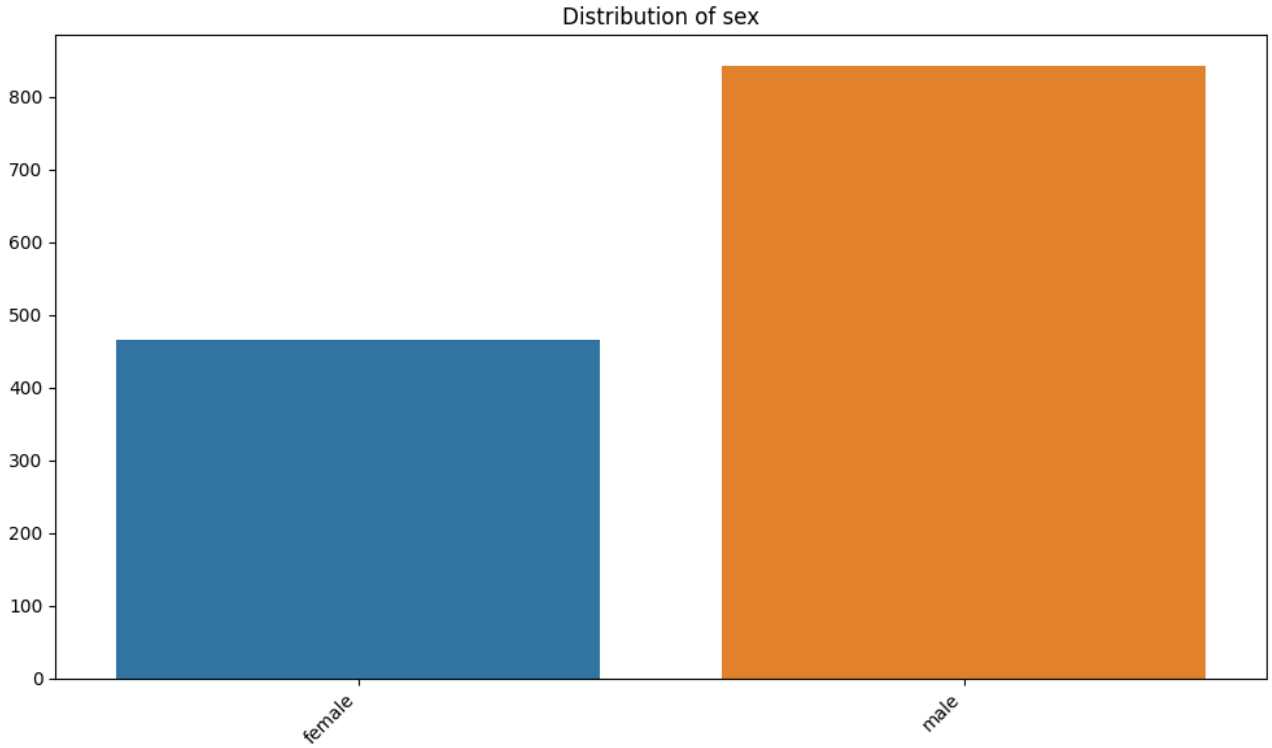

Figure 10: Distribution of sex

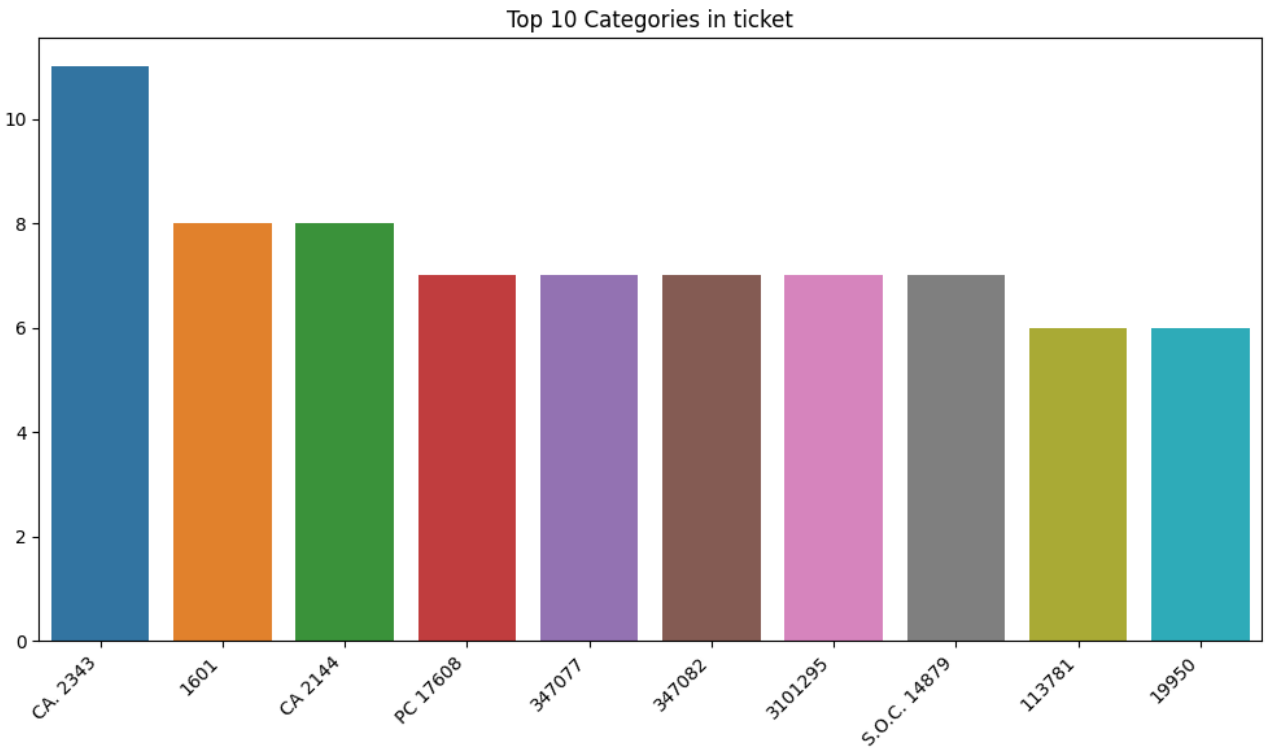

Figure 11: Distribution of ticket

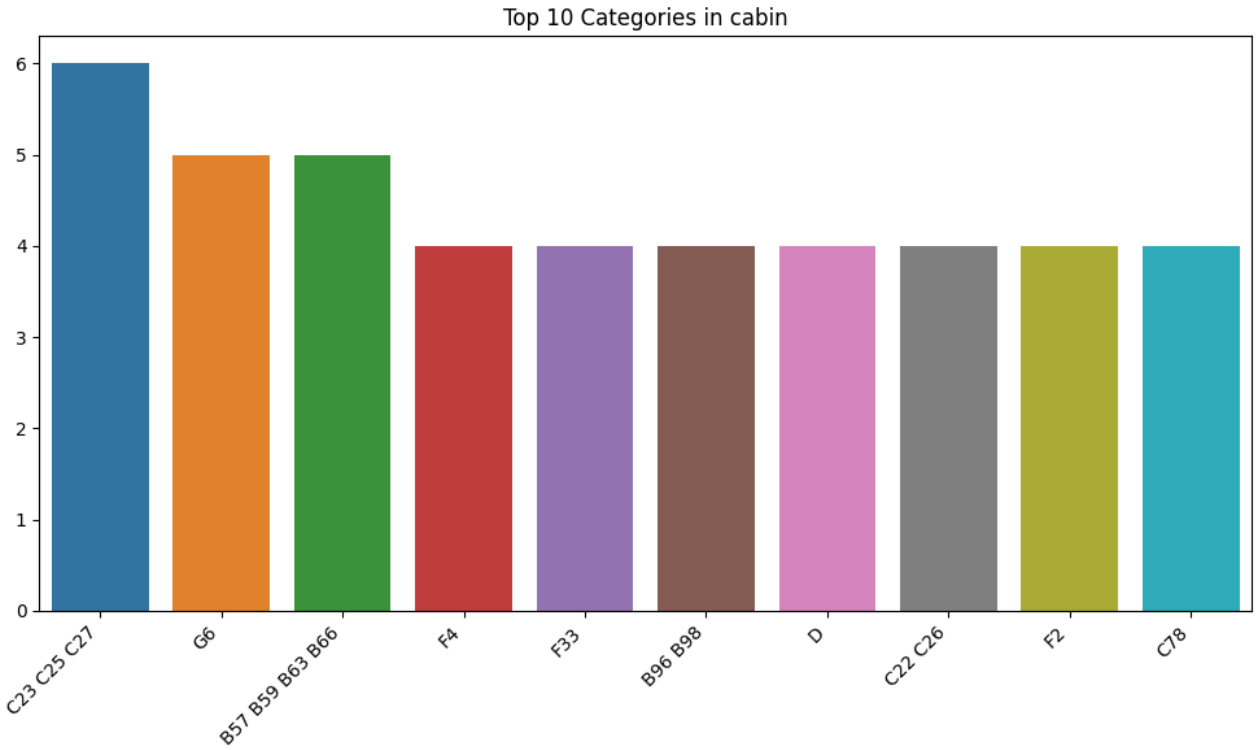

Figure 12: Distribution of cabin

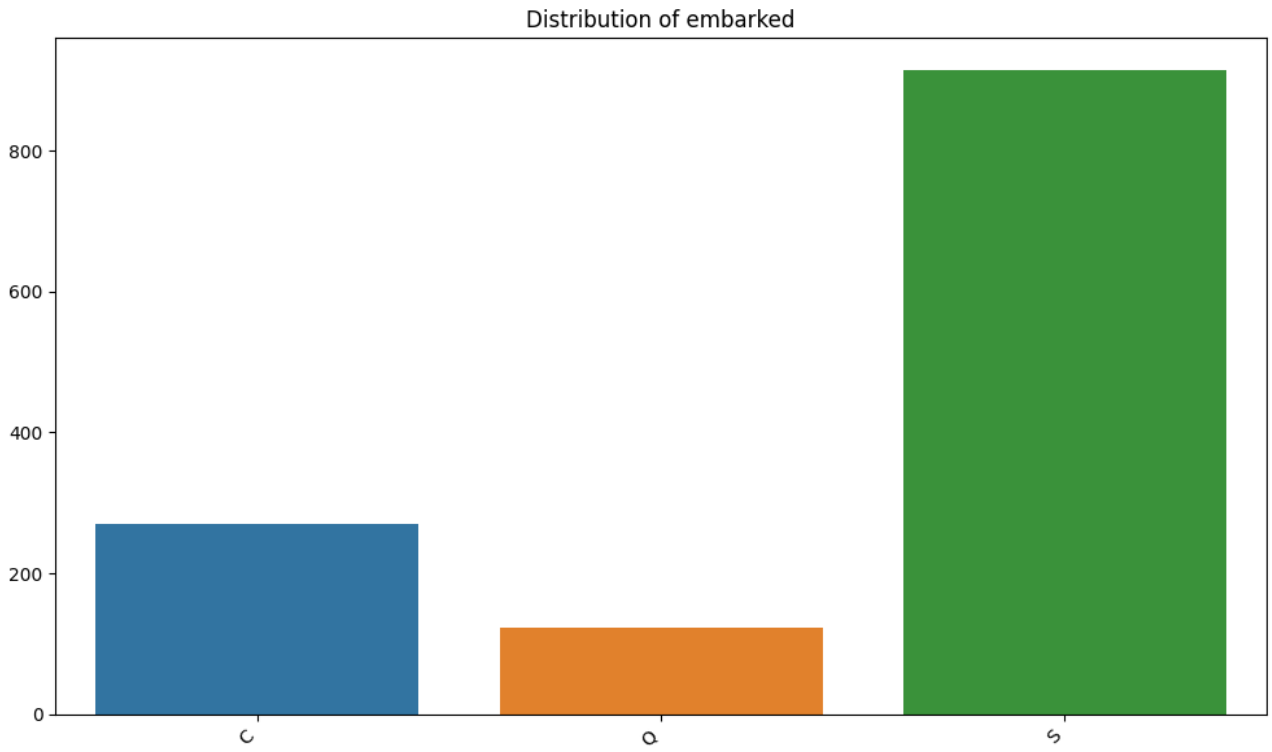

Figure 13: Distribution of embarked

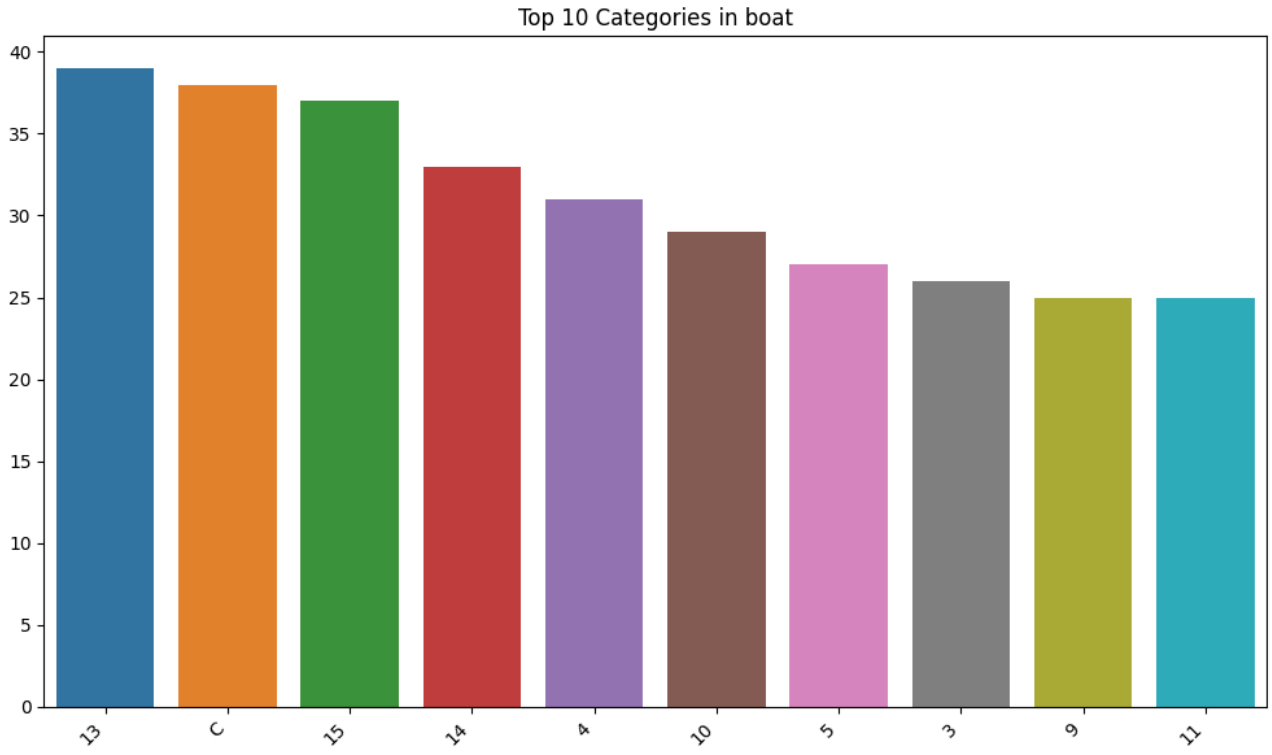

Figure 14: Distribution of boat

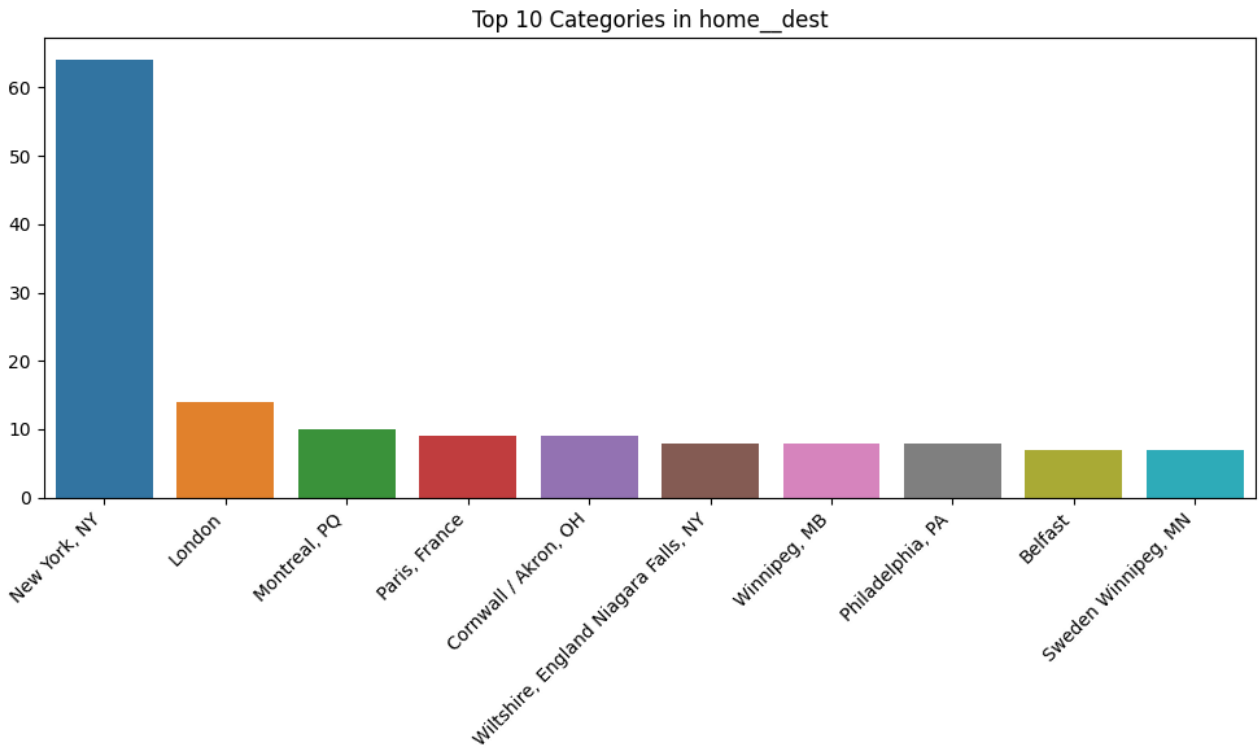

Figure 15: Distribution of home___dest

2.3 Correlation Analysis

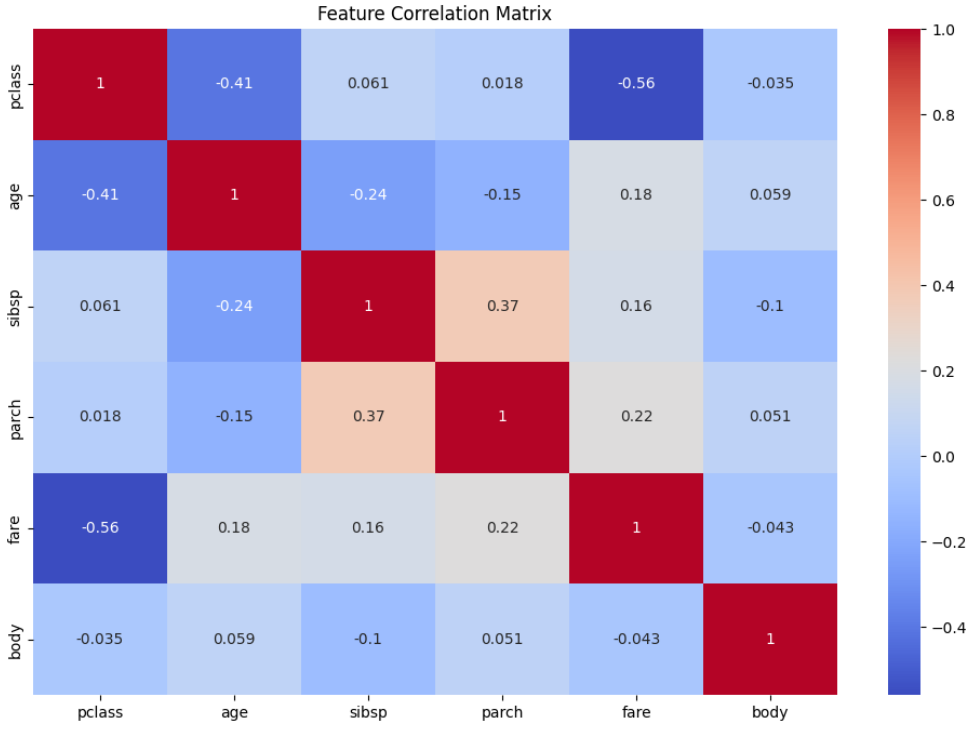

Figure 16: Feature Correlation Matrix

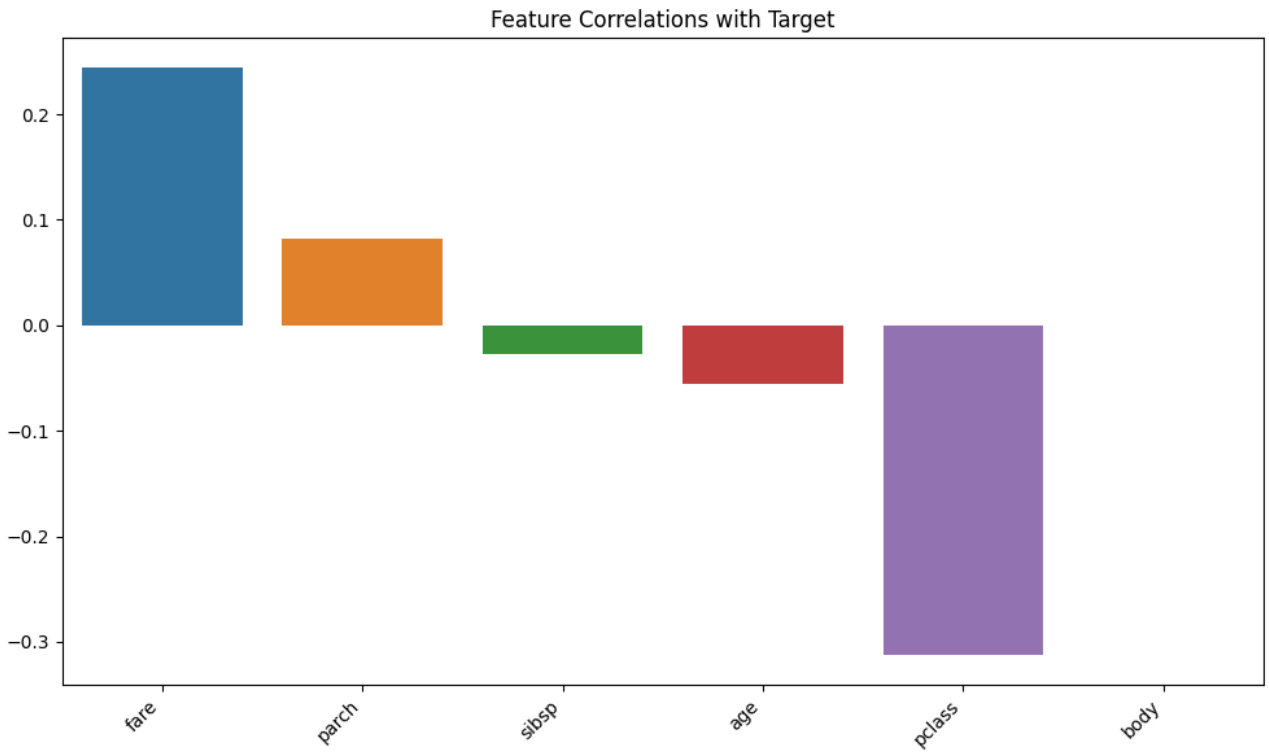

Figure 17: Feature Correlations with Target

3 Model Performance

3.1 Cross-validation Results

5-fold CV Score: 0.9761 (+/- 0.0121)

3.2 Test Set Performance

	precision	recall	f1-score	support
0	0.95	1.00	0.98	144
1	1.00	0.94	0.97	118
accuracy			0.97	262
macro avg	0.98	0.97	0.97	262
weighted avg	0.97	0.97	0.97	262

3.3 Confusion Matrix

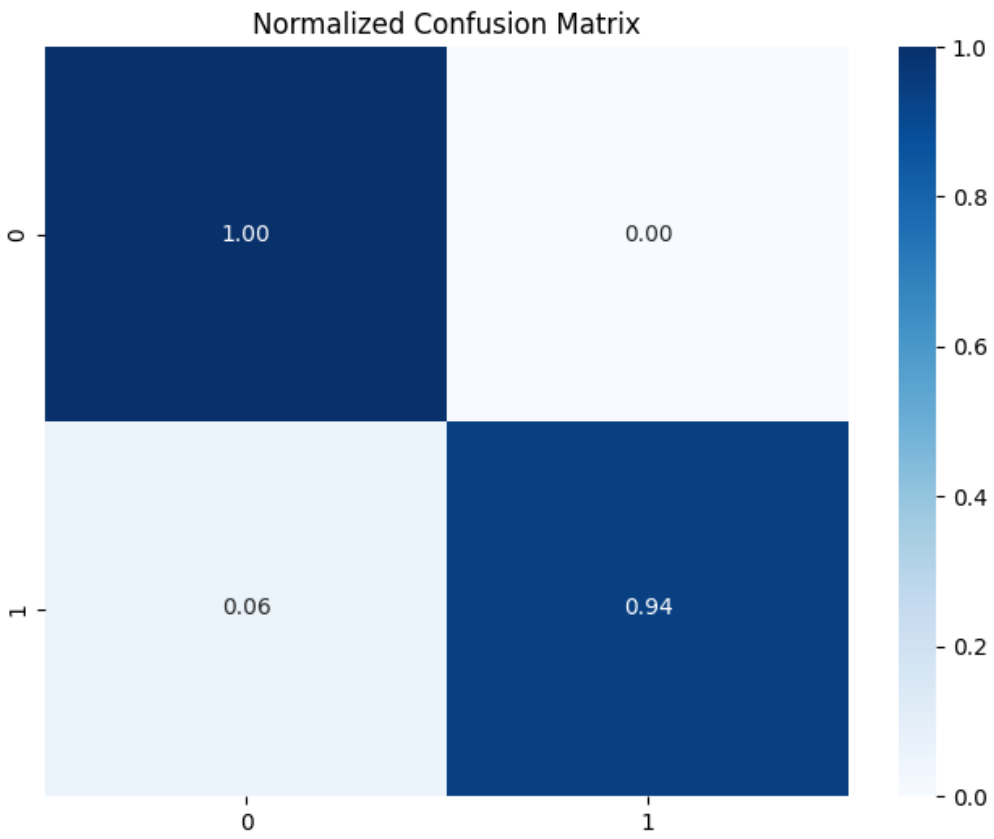

Figure 18: Normalized Confusion Matrix

4 Feature Importance Analysis

Analysis of feature importance in the model.

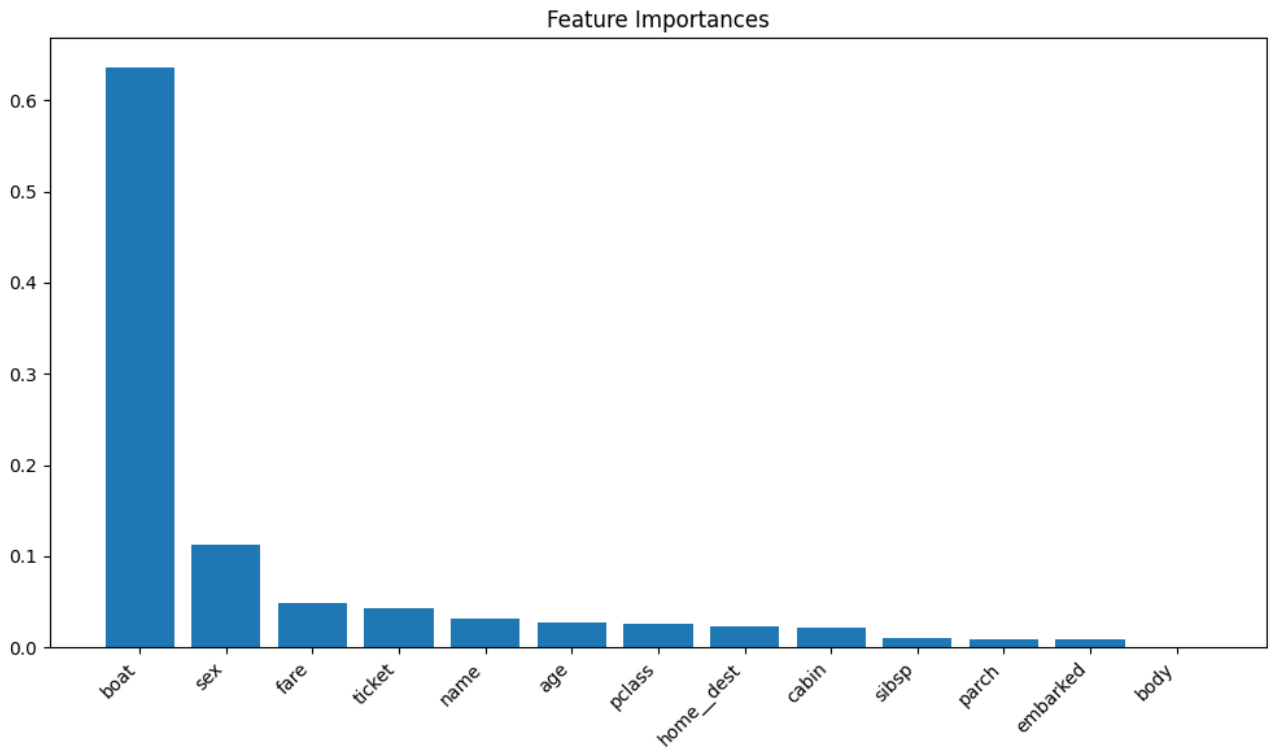

Figure 19: Feature Importance Rankings